



[7. Product and Process Comparisons](#)

[7.4. Comparisons based on data from more than two processes](#)

[7.4.3. Are the means equal?](#)

7.4.3.4. One-way ANOVA calculations

*Formulas
for one-way
ANOVA
hand
calculations*

Although computer programs that do ANOVA calculations now are common, for reference purposes this page describes how to calculate the various entries in an ANOVA table. Remember, the goal is to produce two variances (of treatments and error) and their ratio. The various computational formulas will be shown and applied to the data from the previous [example](#).

*Step 1:
compute
CM*

STEP 1 Compute CM , the correction for the mean.

$$CM = \frac{\left(\sum_{i=1}^3 \sum_{j=1}^5 y_{ij}\right)^2}{N_{total}} = \frac{(\text{Total of all observations})^2}{N_{total}} = \frac{(108.1)^2}{15} = 779.041$$

*Step 2:
compute
SS(Total)*

STEP 2 Compute the total SS .

The total $SS = SS(Total)$ = sum of squares of all observations $- CM$.

$$\begin{aligned} SS(Total) &= \sum_{i=1}^3 \sum_{j=1}^5 y_{ij}^2 - CM \\ &= (6.9)^2 + (5.4)^2 + \dots + (6.9)^2 + (9.3)^2 - CM \\ &= 829.390 - 779.041 = 45.439 \end{aligned}$$

The value of 829.390 is called the "raw" or "uncorrected " sum of squares.

*Step 3:
compute
SST*

STEP 3 Compute SST , the treatment sum of squares.

First we compute the total (sum) for each treatment.

$$T_1 = 6.9 + 5.4 + \dots + 4.0 = 26.7$$

$$T_2 = 8.3 + 6.8 + \dots + 6.5 = 38.6$$

$$T_3 = 8.0 + 10.5 + \dots + 9.3 = 42.8$$

Then,

$$SST = \sum_{i=1}^3 \frac{T_i^2}{n_i} - CM = \frac{(26.7)^2}{5} + \frac{(38.6)^2}{5} + \frac{(42.8)^2}{5} - 779.041 = 27.897.$$

*Step 4:
compute
SSE*

STEP 4 Compute SSE , the error sum of squares.

Here we utilize the property that the treatment sum of squares plus the error sum of squares equals the total sum of squares. Hence,

$$SSE = SS(Total) - SST = 45.349 - 27.897 = 17.45.$$

*Step 5:
Compute
MST,
MSE, and
F*

STEP 5 Compute MST , MSE , and their ratio, F .

MST is the mean square of treatments, MSE is the mean square of error (MSE is also frequently denoted by $\hat{\sigma}_e^2$).

$$MST = \frac{SST}{k - 1} = \frac{27.897}{2} = 13.949$$

$$MSE = \frac{SSE}{N - k} = \frac{17.452}{12} = 1.454$$

where N is the total number of observations and k is the number of treatments. Finally, compute F as

$$F = \frac{MST}{MSE} = 9.59.$$

That is it. These numbers are the quantities that are assembled in the [ANOVA table](#) that was shown previously.